

Advertising in Location-Based Games: An Exploration in *Pokémon GO*

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ABSTRACT

Location-based advertising (LBA), such as billboards and signage, has long been a direct-to-consumer advertising staple. As locative media such as Location-Based Games (LBG) begins to rise in prominence, digital LDA becomes increasingly appealing. In this paper, we explore the impacts of LBA on small businesses in the LBG *Pokémon GO*, a lacuna in the literature. We gather participant experience through a collection of 35 semi-structured interviews with businesses leveraging Niantic's sponsored location LBA. These testimonies indicate (1) participant businesses found LBG advertising to be satisfactory, (2) LBG advertising improves brand recognizability for local commerce, and (3) local community is an important factor for success in LBG advertising. These findings indicate future patterns for integrating local businesses into LBG like *Pokémon GO*, suggesting potential for LBG advertising to assist local businesses.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Collaborative and social computing.**

KEYWORDS

Location-based Games, *Pokémon GO*, COVID-19, Advertising

ACM Reference Format:

John Dunham, Jiangnan Xu, Konstantinos Papangelis, and David Schwartz. 2022. Advertising in Location-Based Games: An Exploration in *Pokémon GO*. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3519663>

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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00
<https://doi.org/10.1145/3491101.3519663>

1 INTRODUCTION

In late 2019 Niantic announced their Sponsored Location program, which enabled businesses to advertise in their Location-Based Games (LBG), particularly *Pokémon GO* (Niantic's flagship title) [17]. While not the first instance of advertising in LBG, this represents one of the first instances of small businesses gaining access to such a service. Moreover, Niantic extended free membership in the program to 1,000 businesses [16], exposing several small businesses to LBG advertising for the first time. This advertising model leverages the hybrid nature of LBG, enhancing the physical location of the business with either a stop or gym (points of interest in *Pokémon GO*), placing local businesses in front of players of the game [18].

The promise of LBG advertising presents an advertising modality wherein the consumer, in being served the ads, directly benefits from the ad in their ludic activity. As these ads are points of interest (PoI) in the game world, they allow players to update their image of their city [14], not only instilling a sense of brand awareness but also a more concrete understanding of the actual location of the business. Furthermore, as this advertising model leverages PoI in LBG, which frequently provides in-game rewards to players, this may engender positive associations with the businesses in player perceptions [19].

However, given this advertising model's relative novelty, there exists a lacuna in the current literature in understanding its impacts. This paper seeks to elucidate this subject matter from the perspective of the businesses, leveraging a collection of 35 interviews from small business owners who participated in the Niantic sponsored location program. In these interviews, we ask businesses (RQ1) what were the critical factors in advertising success and failure in LBG advertising? This work has three key findings: (1) participant businesses found LBG advertising to be satisfactory, (2) LBG advertising improves brand recognizability for local commerce, and (3) local community is an important factor for success in LBG advertising.

We begin the paper with a review of related literature that explores advertising in locative contexts. Then, our methodologies are reviewed, followed by a disclosure of summary information about our participants. The findings of these interviews are then presented in detail. We finish the paper with a discussion of the findings and an exploration of future work.

2 BACKGROUND

As noted in the introductory section, LBG represents an imposition of digital contexts upon physical locations, frequently implemented as mobile apps [4]. In general, this projection of the digital onto the real can be considered as locative media, or media wherein the consumer's physical location has an impact on its presentation. These projections, furthermore, can enhance the imageability [14] of the city as they recontextualize the space occupied by the users [18]. While *Pokémon GO* is one of the more popular pieces of locative media at present, earning over one billion US dollars in 2020 alone [8], and the probe leveraged in this study, it is not the only form locative media extant.

Foursquare, a social media platform from 2009, leveraged locative technology to create a social network wherein users could claim digital ownership over physical locations [12]. This social network unveiled several critical concepts at the heart of modern locative media, particularly territoriality. In Foursquare, users could check-in at locations with the capacity of becoming the "mayor," effectively the digital owner, of the place if they had visited it the most in a 60 day period [12]. Interestingly, investigating user experience of this platform indicated that check-ins were largely performative and conditional for the users as noted by Lindqvist et al. in a study of Foursquare [12], supported by a follow-up study by Papangelis et al. leveraging a separate locative media probe [18].

The finding of users' performative self is crucial, as it indicates that users of locative media mediate their presentation of self when engaging with it. User mediation of digital self parallels Goffman's dramaturgical lens of human behavior, wherein the individual has a public and private persona [11]. Returning to locative media, the existence of a performative digital self suggests that the norms of the local community impact how players interact with points of interest (PoI)

Researchers have studied the nature of socialization in *Pokémon GO* and its impact on the game at length. Prior work has shown local communities to exhibit strong communication and social interaction between players of the game [10, 13]. Moreover, before the COVID-19 pandemic, contextual dynamics [1] between players resulted in a great deal of in-person interaction [5] during in-game events such as community days and raiding. Community days are social events wherein pokémon spawn more frequently, encouraging players to go out and play the game (in pre-pandemic contexts) [15]. Raiding is a communal activity wherein multiple players come together to challenge enemies in-game that must be defeated with the help of other players [15].

These community dynamics and player tendencies for territoriality are vital to Niantic's objective of making PoI a means of advertising. Moreover, businesses have recently been given access to LBG advertising through Niantic's sponsored location program, opening research space into the impacts of the technology [17]. A sponsored location allows the business to create and modify their business's presence in *Pokémon GO*, as well as sponsor in-game events (currently disabled due to COVID-19) and start in-game promotions. For 1 US Dollar a day, the business is given a pokéstop (the lower tier of PoI), and at 2 US Dollars, the PoI becomes a gym; both tiers provide the business with analytics about their location. Business owners can control their digital presence directly through

Niantic's program, allowing them to brand their location as they see fit in the context of LBG.

In light of the outbreak of the COVID-19 pandemic, shortly after launching the sponsored location program, Niantic also launched the "Local Business Recovery Initiative" [16], which allowed communities to nominate small businesses for inclusion in the sponsored location program for a year free. Most of our participants are beneficiaries of this program, classifying as Niantic's definition of a small business (five or fewer storefronts) [16]. Naturally, this inherently entwines the sponsored location program with the community, establishing community engagement as a vital component to this means of advertising.

Locative media augments the Location-Based Advertising (LBA) of old (e.g., billboards and signage), allowing the advertisements to address the consumer directly through direct marketing based on location in real-time [3]. Work conducted by Xu et al. [21] indicates that mobile, multimedia, LBA leads to more favorable responses from consumers than traditional advertising. This work indicates that this distinction is borne from a perception of increased value in the mobile LBA, as the information is generally more relevant to the consumer. Moreover, perception of value leads to better advertising outcomes in general, as noted in Wang and Sun [19]. As a result, mobile LBA with location-aware services appears to have promise and value to businesses in the general case. However, to date, most research in this sphere focuses on targeted push-based advertising [3].

Limited exploration has been conducted on marketing potential in LBG: Wu and Stillwell conducted a survey (N=352) of *Pokémon GO* players finding that spatial presence positively influenced user attitudes towards and intention to visit sponsored locations [20]. Yet, the majority of research in this sphere to date has focused on targeted push-based advertising [3]. While advertising through LBG such as *Pokémon GO* is not an entirely new frontier, there still exists a marked gap in the literature on this subject, and one we address with this research. This paper, therefore, seeks to reduce the knowledge gap of LBA advertising through LBG impacts, achieved through a set of qualitative interviews with small business owners enrolled in the Niantic sponsored location program. In the following section, we will outline the methodological approach underpinning this research.

3 METHODOLOGY

We conducted a series of semi-structured interviews with participants of Niantic's sponsored location program. The first author acquired participants with the assistance of the operations department of Niantic. Researchers determined interview eligibility on three factors: (1) the business participated in either the paid or free version of the sponsored location program within the last year, (2) they self-identified as a small to medium business, and (3) had previously consented to participate in studies of the program efficacy with Niantic. Our participants were offered no additional incentives for participating in the interview.

We enabled registration for the interview through the usage of the external online calendar tool Calendly [7]. Researchers conducted interviews over 45 minutes periods scheduled by participants. The calendar tool asked participants for names, email addresses, and business names when registering. The registration also presented the informed consent document to be approved by the participant or flag the interview for a deeper discussion of the participant's rights before conducting the discussion. Pseudonyms were then applied to all participants to address privacy concerns.

In total, researchers invited 166 businesses to participate in these interviews. Sixty-three businesses consented to participate in the study; however, only 36 completed their interviews, the remaining 27 did not attend the scheduled Zoom meeting. Of the 27 businesses that failed to complete their interview: one dropped out, deciding the discussion would be pointless, citing low interaction with their PoI. In another case, language barriers prevented the completion of the interview. Finally, we discarded one of the 36 completed interviews as the content diverged too far from the study's stated goals.

Interviews began with a summary of the informed consent document, outlining their rights. After affirming consent, researchers asked if participants were comfortable with Zoom [22] recording and auto transcribing their interviews. All participants assented to being recorded, with each interview taking approximately 30 minutes to complete.

Structurally the interviews were centered along with three core categories: (1) demographic information surrounding the business, (2) reasons for joining the program and broad outcomes, and (3) direct questions about the features afforded to sponsored locations. Collaboration between authors of this article and Niantic's operations team generated a set of 21 questions centered around these three core categorizations, with possible clarifying questions provided. Researchers designed questions and possible follow-ups as clarifying questions, for example, "Do you find the sponsored location to be valuable to your business operation?" was followed up by "What would you be willing to pay for this service?" The wording of these questions was deliberately open to allow the interviewer to modify the language to match the conversation flow to elicit contextual information surrounding the participant's response.

Following the execution of the interviews, the first and second authors of this paper conducted independent inductive thematic analysis [2, 6] of 5 randomly selected interviews (using the Ubuntu *RANDOM* construct). In addition to the interviews themselves, author analysis included brief discussions immediately following interviews. We then generated a resulting thematic framework from combining these observations in a meeting discussing independent findings. The researchers leveraged this resultant thematic framework independently on the remaining interviews, iteratively discussing the thematic framework through regular communication. The core emergent themes include (1) local community impacts, (2) business enrichment through *Pokémon GO*, (3) pain points in running a location, (4) business type and engagement model, and (5) participant satisfaction with LBG advertising. This work-in-progress paper highlights the impact of the local community on LBG advertising and explores participant satisfaction as mediated by this theme.

Demographically our participants came from 4 countries: the United States (19/35), Canada (11/35), Mexico (4/35), and the UK (1/35). The average number of employees ranges from 2 (typically run by couples or partners) to 100 (a local entertainment center), with the median number of employees being 6. While the largest businesses, in terms of payroll, were found in the United States, the surveyed Canadian businesses were more likely to have an employee count greater than 10. 5 of the 11 observed Canadian businesses have more than ten employees to the United States rate of 4 to 19. Neither Mexico nor the UK had an employee count of more than 10.

4 FINDINGS

In terms of types of businesses questioned, we've separated them into five categories for analysis: friendly local game stores (FLGS) (10/35), entertainment venues (6/35), retail (6/35), restaurants (9/35), and professional services (4/35). FLGS are locations where customers purchase products and engage in ludic activities, such as playing trading card games (TCGs), board games, or wargaming. These locations have a high degree of synergy with *Pokémon GO*, nearly every business of this classification (9/10) considered the sponsored location program, as explicitly noted by B12, as "a kind of cross-promotion." Entertainment venues ranged from bowling alleys to museums, while Professional services could range from health services to tax services. Retail venues were similarly diverse, although characterized by a lack of integration in their products with the *Pokémon* franchise in this study. Restaurants were both sit-down and take-out, selling a wide variety of foods.

We also found it helpful to assess the participating businesses as having transitory or sustained customer engagement. Transitory engagement represents businesses that value the scale of throughput. A transitory business is generally affiliated with retail, professional services, and take-out restaurants, as these businesses have less obvious value in customers staying at the location for extended periods. Conversely, the sustained engagement business favors keeping customers at the location, as it directly leads to the customer spending more money. We identified FLGS, entertainment venues, and sit-down restaurants as fitting this interaction model. B28 codifies sustained engagement: "the idea is to keep them long enough [that] .. they buy a soda or something ... and I make a couple of bucks off of the situation." This distinction appears to impact the services most desired by these businesses, with sustained business types wanting a more feature-rich experience than their transitory counterparts.

4.1 Location, Engagement, and Foot-Traffic

A number of the surveyed businesses (9/35) exist in areas with low foot traffic where car culture was more common, meaning customers were more likely to arrive by car. Frequently these businesses described being in more secluded locations or locations on highways. While business owners observe car play (despite warnings from the *Pokémon GO* app), such locations aren't ideal places to engage in LBG play. However, of the nine participants describing their location as getting more car traffic, six of them surprisingly reported foot traffic increases. B21 offered player curiosity as one potential explanation for this disconnect: "I know for me and other

players if you see a new gym you just say 'Oh let's just walk by it!' That brings people in, and [they] check you out. I think it's very helpful [to] business owners."

B10, on the other hand, found the appeal to be due to the proximity of their location to the play community: "[our region] is spread out, it's not like a city, so [the Gym] made it more centralized ... [rather than] going 10 or 15 minutes up the road to the municipal building, or a new town. So me putting the gym there, I think kind of helped some people that don't want to drive that far." This explanation appears to be more tied to rural locations than urban ones in which there's a greater average distance between points of interest. However, the most striking rationale for the increase came from B06, who described a local community that practiced "small business local support ... [by being] more engaged in their local areas." B06, however, also appeared to be more active in reaching out to the LBG community: "we could see customers who are playing *Pokémon GO*, we start talking to them, ... [then] they bring a friend ... from out of town [for] community day. We've been known as the place to come [to play *Pokémon GO*]." B06's success provides evidence that business engagement with the LBG community begets engagement from the community. Moreover, some participants (12/35) wanted to facilitate these engagement conversations by leveraging physical promotions such as window clings. As B11 notes: "We use that poster, and then people know ... even if they're just playing *Pokémon GO* if they want to escape the elements [they can] come in and play, that they have a place."

The three remaining cases which failed to see foot traffic increases in locations with car culture appear to be more tied to the behaviors of the local players. These local players, according to B33, were more focused on raiding and considered the sponsored gym as just another stop on routes that are more or less optimized to maximize gains from raiding: "I'm just assuming that the *Pokémon GO* folks are just the ... diehard [ones] that are driving around ... to get the benefit of the special raid and then move on [saying] 'I don't need to go into the game store and buy a Pokémon pack ... I'm not into that.'" The disconnect between players using the stop and patronizing the location further supports the notion that the local *Pokémon GO* community has a bearing on the efficacy of the program.

Overall, however, 20 of 35 of the surveyed businesses reported increases in their foot traffic. To business owners reporting this increase, they generally appeared to have the belief that the program was doing *something* to draw attention to their store. Many of the business owners in this categorization offer explanations similar to B21's above. Furthermore, 26 of the participants thought that the Sponsored Location served as a means of brand awareness, with 15 of those participants also reporting increases in foot traffic. Participants perceived the in-game location as a digital billboard for their business. Bookstore B23, for example, joined the program "so [they] can get some neighborhood recognition, so people realize 'oh there's a bookstore over here.' People that play *Pokémon GO* might not notice us because we are kind of in an out of the way location." B23 also reported that this perception was effective for their business: "[We got] a different clientele to notice us and stop in, [or] at least pull into the parking lot [to] do the gym or a raid ... [Some of] these people came into the store to look around." B04, an FLGS, mirrored this sentiment: "Many people have come [to the

store], and they tell me that they [found] us because they [saw] the stop of the business and in the small description it says we're a hobby store." These two stores operate on separate engagement models in our framework. The bookstore expects more transitory interactions from their customers, while the FLGS seeks sustained customer interactions, suggesting a business agnostic quality for the value of the locations as billboards.

4.2 General Satisfaction and Overwhelming Recommendation

Despite apparent merit in the program for brand awareness and increases in foot traffic, some participants (8/35) who reported increases in foot traffic also appeared to be unsure of the program's efficacy. Participants' lack of certainty was a broader trend, with 19 of the businesses reporting uncertainty on the actual efficacy of the program. This uncertainty primarily appears to be related to the local community not communicating *Pokémon GO* as a factor in patronizing the business. A lack of interaction over the game with businesses is likely rooted in communal norms as noted by B13: "Players are really trained, not to bug the businesses there's a gym attached to, ... I think the entire community is trained, not to disrupt."

Niantic does offer some tools to counteract this attribute of local communities, but participants report generally mixed results in leveraging these utilities. Niantic gave participant businesses access to an administration interface that afforded them basic control over how their location was presented in-game: location photo, description, and analytics. Despite the tools and introduction in the startup email, a plurality of participants (17/35) reported ignorance of the interface functionality. B16 offers the most common explanation for this knowledge gap: "I got the initial email about how to set it up, ... but I've kind of forgotten, and I haven't been asked about it until now." B16 suggested reminder emails to business owners when asked for a potential solution.

Despite this pain point, an overwhelming majority of participants (29/35) would or had recommended the program to other small and medium businesses. Some recommendations naturally aligned with the participants' experiences, as all participants who reported a positive experience (20/35) with the program recommended it, citing their own experiences. The remaining nine participants willing to recommend the program had mixed feelings about their experience (8/35) or were generally not satisfied (1/35). In the case of the participant who reported low satisfaction, they had previously introduced the program to other business community members earlier in their tenure with the program. The remaining participants' reasoning to recommend the program fell to two key points: the location of the business and the industry of the business. B03 indicated that the program would benefit businesses where "foot traffic is high, [as] foot traffic is everything. I would recommend it to someone in a large urban center." B11 recommended the game to "businesses [with] similar overlap [like] arcades, comic book stores. Places that already sell Pokémon toys and products. The remaining six participants offered that they would conditionally recommend the program. Recommendations in this manner generally focused on the business type, targeting businesses like "gaming cafes" (B09) or a "gaming house" (B14).

5 DISCUSSION

While not wholly universal, the key benefit of LBG advertising reported by our participants appears to be its capacity to foster brand awareness and increase the imageability of the business in players [14]. The PoI acts as a digital anchor or billboard for the business imparting locative information and brand awareness. Interestingly, as the advertisement takes the form of an in-game PoI, the ad is not relegated to the periphery; instead, there exists a strong mechanical integration between the ad and gameplay. As the advertisement provides direct in-game benefits to the player, it may be less irritating than traditional multimedia advertisements injected into the game (e.g., banner ads and full-screen ads). Moreover, the integration of the business location into the game is diegetic, differing minimally from the non-sponsored locations. Participant responses appear to support the program's value to players, regardless of conversion to customers. More importantly, the PoI appeared to influence foot traffic to the participant businesses, with 20 of the 35 participants reporting increases in foot traffic, indicating a modest success in the PoIs function as a digital billboard. This reasoning for successes in the program is in line with existing research linking positive perceptions of an advertisement by consumers resulting in favorable business results [19].

Yet, we are left to wonder what happens in the remaining cases where there is no noticeable increase in foot traffic. We turn to the locations with heightened car cultures to better understand such cases. The split between foot traffic increases and no reported increases appears to align with the broader trend (three increases to one neutral). These foot traffic trends suggest that the business's physical location doesn't appear to have significant sway over the efficacy of LBG advertising in the context of foot traffic. Instead, there appears to be a deeper relationship between the local player community of the region in which the PoI and business reside. As noted above, community support for local businesses within the LBG community was present in the locations of some of the participants. Naturally, such community support led to increased foot traffic experienced by the associated businesses.

In contrast, participants described local communities which avoided interfering with local businesses in the execution of play due to previous experiences and community norms. This play model would manifest itself as though the players were opportunistic, only focused on in-game objectives. In these communities, foot traffic saw a minimal change, as there was a minimal incentive in many cases to actively interact with the physical location. The low impact on foot traffic may also be an extension of territoriality, albeit not so much as players declaring ownership [12, 18], but the tensions between physical and digital ownership. While players can attain a sense of ownership over locations in LBG, this ownership is ephemeral and contextual. Physical ownership (e.g., a business or a home), in contrast, appears to be more concrete to players and takes precedence over the digital world. The dynamic between digital and physical ownership may result in players' perceptions of space becoming muddled, making entering territories with these unclear dynamics unappealing or awkward.

Regardless of the mechanism for this lack of increase, these findings indicate that to ensure positive increases in foot traffic, LBG advertisements in the form of PoIs should incentivize or encourage

interaction with the physical location. In actively inviting players to occupy the space, the PoI itself can remove ambiguity in the space and more clearly convey the intentions of the business. Direct interaction from the business with the community itself also appears to remove spatial ambiguity. In bringing *Pokémon GO* into the real world, the business can reduce player anxieties about playing the game in retail spaces. Some participants (12/35) even expressed some interest in physical advertising of the stop to facilitate conveying the validity of the location as a place to play the game. Interestingly, there appear to be some mechanisms present today to help facilitate this type of outreach; however, few participants leveraged this tool in the interview pool.

Despite the apparent shortcomings of the current iteration of the LBG advertising, participants were highly positive about their experiences, as nearly every participant was willing to or had recommended the program. For some, this was a genuine satisfaction with the program, as it achieved their expected results. More interestingly, a substantial population of participants who reported mixed or negative results still would be willing to recommend the program. This support for sponsored locations seemed to be focused on the promise of LBG advertising, recognizing its potential as a digital billboard, particularly to overlapping businesses. This program represented an untapped opportunity to our participants, rife with potentially novel applications. Moreover, our participants were nearly universally familiar with *Pokémon GO* and players themselves, offering a unique perspective on the technique. Pre-existing experiences may be responsible for this overwhelming support; however, it still indicates an advertising avenue teeming with enthusiasm.

6 CONCLUSION

This work represents fledgling steps of research into LBG advertising. Methodologically, we focused on the business owners to determine the program's value, perceived or otherwise. In starting with the businesses themselves, we can ascertain the potential for the advertising technique. Pursuant to this objective, researchers conducted a collection of semi-structured interviews asking the question (RQ1) what were the critical factors in advertising success and failure in LBG advertising? While this work only begins to scratch the surface of this question, we did find three key components of LBG advertising: (1) participant businesses found LBG advertising to be generally satisfactory, (2) LBG advertising appears to have an impact on brand recognizability for local commerce, and (3) the local LBG community's norms have a direct impact on successes in LBG advertising. Moreover, the program appears to have promising potential to support local businesses, increasing awareness of their brands.

Unsurprisingly, socialization, centered on the local community, appears to be at the heart of LBG advertising, as it is to LBGs themselves [9]. These findings only serve to encourage new questions: where do owners derive their satisfaction with LBG advertising? How do businesses run LBG advertising campaigns, and what actions yield maximum results? Does LBG Advertising have additional benefits for locations beyond brand awareness and imageability? Researchers must further assess business owners' experiences with LBG advertising to answer these questions.

In terms of limitations, our work was limited to English-speaking participants, preventing interviews with non-English speaking participants in the sponsored location program. Additionally, while we directly pulled our recruitment pool from the list of businesses in the sponsored location program, respondents are likely biased towards more extreme outlooks on their experiences. Future work should be conducted in non-English speaking regions, while the participants must be considered through this lens when assessing the present research findings.

We seek to open the discussion on this novel advertising modality through this work. While there exist explorations of LBA, this new sub-modality represents new opportunities with differing requirements. Further research in this space must target not only the experiences of the businesses but the impact of LBG advertising on the players.

ACKNOWLEDGMENTS

Niantic has funded this work through the Niantic X RIT Geo Games and Media Research Lab at the Rochester Institute of Technology.

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